

lift acting on an incremental spanwise element of the wing:

$$l(y) = \rho_{\infty} U_{\infty} \Gamma(y)$$

Total lift

$$L = \int_{-s}^{+s} \rho_{\infty} U_{\infty} \Gamma(y) dy$$

The total vortex drag

$$D_v = - \int_{-s}^{+s} \rho_{\infty} w(y) \Gamma(y) dy$$

Induced velocity at any point y_1

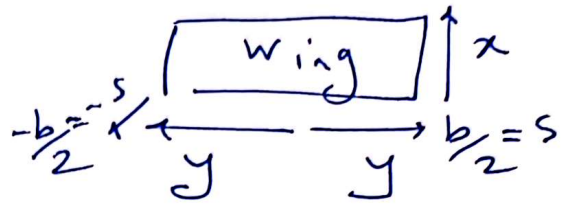
$$w_{y1} = + \frac{1}{4\pi} \int_{-s}^{+s} \frac{d\Gamma/dy}{y - y_1} dy$$

Downwash angle

$$\epsilon = \tan^{-1} \left(\frac{w_{y1}}{U_{\infty}} \right) \approx - \frac{w_{y1}}{U_{\infty}}$$

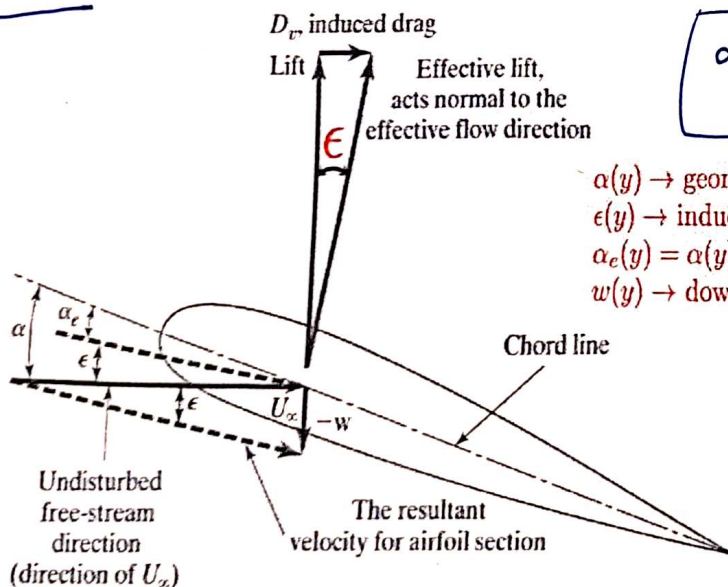
Effective angle of attack of the airfoil

$$\alpha_e = \alpha - \epsilon$$



$$\frac{\Gamma}{4\pi R} = \int \frac{\frac{d\Gamma}{dy} dy}{4\pi (y - y_1)}$$

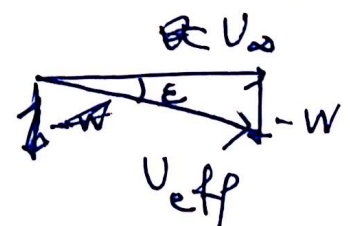
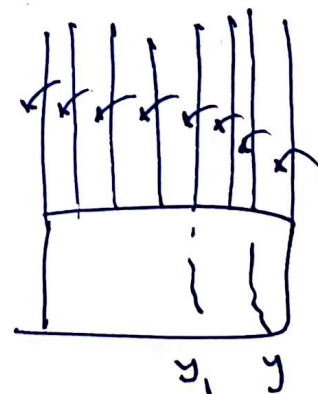
$-\frac{b}{2} < y < \frac{b}{2}$



- Chord line of the airfoil
- Effective flow direction
- Undisturbed free-stream direction

$$\alpha_e = \alpha - \epsilon$$

- $\alpha(y) \rightarrow$ geometric AOA
- $\epsilon(y) \rightarrow$ induced or downwash AOA
- $\alpha_e(y) = \alpha(y) - \epsilon(y) \rightarrow$ effective AOA
- $w(y) \rightarrow$ downwash velocity



$$-1 < \frac{y}{s} < 1$$

$$0 < \phi < \pi$$

Case of Elliptic Spanwise Circulation Distribution

Elliptic circulation distribution

$$\Gamma(y) = \Gamma_0 \sqrt{1 - \left(\frac{y}{s}\right)^2} \rightarrow s = \frac{b}{2}$$

Induced downwash velocity

$$w_{y1} = w(y) = -\frac{\Gamma_0}{4s}$$

Total lift for the wing

$$L = \rho_{\infty} U_{\infty} \Gamma_0 s \frac{\pi}{2} = \left[\frac{\pi}{4} b \rho_{\infty} U_{\infty} \Gamma_0 \right] \quad L = \int_{-s}^s \rho_{\infty} U_{\infty} \Gamma(y) dy$$

Lift coefficient

$$C_L = \frac{L}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 S} = \frac{\pi b \Gamma_0}{2 U_{\infty} S}$$

Downwash angle

$$\epsilon = \frac{w}{U_{\infty}} = + \frac{\Gamma_0}{4 U_{\infty} s} = \frac{\Gamma_0}{4 U_{\infty} \frac{b}{2}} \quad \epsilon = \frac{\Gamma_0}{4 U_{\infty} \frac{b}{2}} = \frac{2 C_L U_{\infty} S}{4 U_{\infty} \frac{b}{2} \pi b} = \frac{C_L}{\pi AR}$$

Total vortex (or induced) drag for the wing.

$$D_v = \frac{\rho_{\infty} \Gamma_0^2}{4s} \int_0^{\pi} \sqrt{1 - \cos^2 \phi} \sin \phi d\phi \quad dy = s \sin \phi d\phi$$

$$= \frac{\pi}{8} \rho_{\infty} \Gamma_0^2 \quad \left\{ \frac{y}{s} = \cos \phi \right.$$

Drag coefficient

$$C_{Dv} = \frac{D_v}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 S} = \frac{\pi \Gamma_0^2}{4 U_{\infty}^2 S} \quad C_{Dv} = \frac{C_L^2}{\pi AR}$$

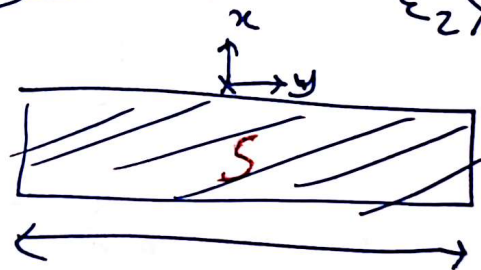
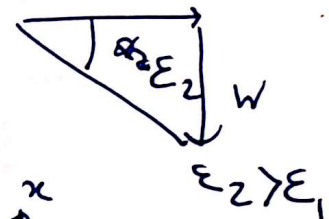
Geometric angle of attack for an elliptic lift distribution

$$\alpha = \alpha_e + \frac{C_L}{\pi AR}$$

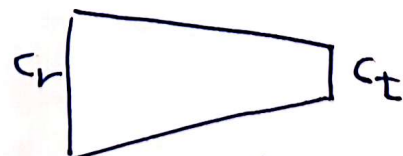
Lift-curve slope

$$a = C_{L_{\alpha}} = \frac{a_0}{1 + \frac{a_0}{\pi AR}} = \frac{C_{L_{\alpha}}}{1 + \frac{C_{L_{\alpha}}}{\pi AR}}$$

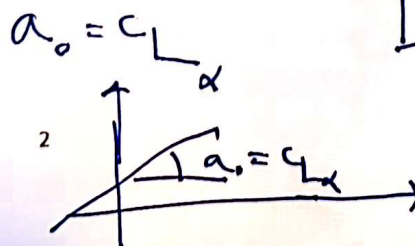
a_0 : Airfoil lift slope



$$AR = \frac{b^2}{S}$$



$$\lambda = \frac{c_t}{c_r} \leq 1$$



General Spanwise Circulation Distribution

$$\Gamma(\phi) = 4sU_{\infty} \sum_1^N A_n \sin n\phi$$

→ Sin Fourier Series

$$\Gamma(\phi) = 4sU_{\infty}(A_1 \sin \phi + A_3 \sin 3\phi + A_5 \sin 5\phi + \dots)$$

$$\frac{y}{s} = -\cos \phi$$

The section lift force

$$l(\phi) = \rho_{\infty} U_{\infty} \Gamma(\phi) = 4\rho_{\infty} U_{\infty}^2 s \sum_1^N A_n \sin n\phi$$

Section lift coefficient

$$C_l(\phi) = \frac{\rho_{\infty} U_{\infty} \Gamma(\phi)}{\frac{1}{2} \rho_{\infty} U_{\infty}^2 c} = \frac{2\Gamma(\phi)}{U_{\infty} c}$$

$$C_l = \left(\frac{dC_l}{d\alpha} \right)_0 (\alpha_e - \alpha_{0l})$$

$$\alpha_0 (\alpha_e - \alpha_{0l})$$

Downwash angle

$$\varepsilon = -\frac{w}{U_{\infty}} = \frac{\sum n A_n \sin n\phi}{\sin \phi}$$

General Prandtl Equation

$$\mu(\alpha - \alpha_{0l}) \sin \phi = \sum A_n \sin n\phi (\mu n + \sin \phi) \quad (*)$$

$$\mu = ca_0 / 8s$$

Solve this equation for n-sections of the wing to get $A_1, \dots, A_n$

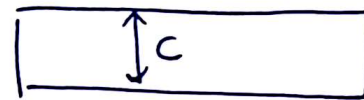
Lift on the Wing

$$L = \int_{-s}^{+s} \rho_{\infty} U_{\infty} \Gamma(y) dy = \int_0^{\pi} \rho_{\infty} U_{\infty} s \Gamma(\phi) \sin \phi d\phi$$

$$L = 4\rho_{\infty} U_{\infty}^2 s^2 \int_0^{\pi} \sum A_n \sin n\phi \sin \phi d\phi$$

$$L = (4s^2) \left(\frac{1}{2} \rho_{\infty} U_{\infty}^2 \right) A_1 \pi = C_L \left(\frac{1}{2} \rho_{\infty} U_{\infty}^2 \right) (S)$$

7.2



rectangular wing → $s = cte$

c is cte

No aerodynamic twist
airfoils are the same
 α_0 is cte

A_n is unknown

$$\mu = \frac{c a_0}{8s} = cte$$

So write (*)

at various sections



$$\frac{c_t}{c_r} = 1 \quad \text{follow Ex. 7.2}$$

in book

to get μ

$$\Gamma \propto \sum A_n \sin n\phi$$

Just A_1 appears in lift

Wing lift coefficient

$$C_L = A_1 \pi AR$$

Vortex-Induced Drag

$$D_v = - \int_{-\pi}^{+\pi} \rho_{\infty} \omega \Gamma dy = 4 \rho_{\infty} s^2 U_{\infty}^2 \int_0^{\pi} \sum n A_n \sin n\phi \sum A_n \sin n\phi d\phi$$

Coefficient for the vortex-induced drag

$$C_{Dv} = \pi \cdot AR \sum n A_n^2$$

$$C_{Dv} = \frac{C_L^2}{\pi \cdot AR} \sum n \left(\frac{A_n}{A_1} \right)^2$$

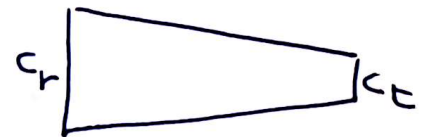
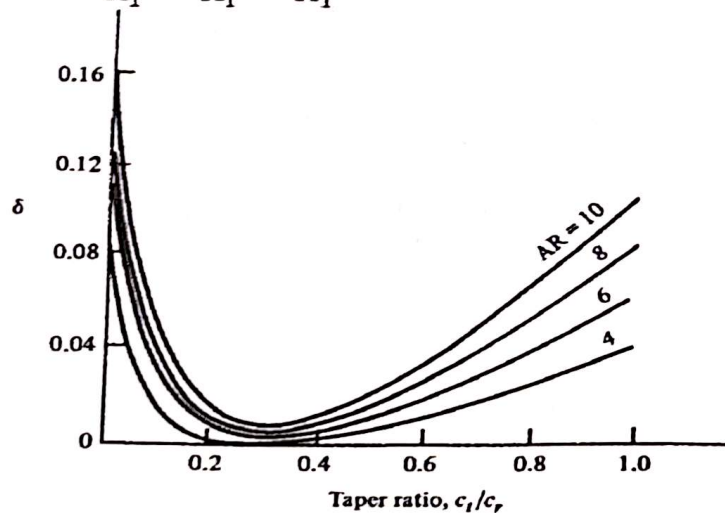
$$C_{Dv} = \frac{C_L^2}{\pi AR} (1 + \delta) = \frac{C_L^2}{\pi e AR}$$

Span efficiency factor

$$e = 1/(1 + \delta)$$

$$\delta = \frac{3A_3^2}{A_1^2} + \frac{5A_5^2}{A_1^2} + \frac{7A_7^2}{A_1^2} + \dots$$

$$e = \frac{1}{1 + \delta}$$



$$\begin{cases} \frac{c_t}{c_r} = 0.3 \\ AR = 4 \end{cases}$$

You can read δ from this chart.

Lift curve slope for a general wing usually assumes $\delta = \tau$

$$a = \frac{a_0}{1 + (a_0/\pi AR)(1 + \tau)}$$

lift slope of airfoil

works similar to elliptic wing

lift slope of wing $AR \equiv \frac{b^2}{S}$

Taper ratio $\equiv \frac{c_t}{c_r}$

$$\mu = \frac{ca_0}{4b} = \frac{5 \times 2 \times \pi}{4 \times 40} = 0.19625$$

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$$b^2 = b \times c = 200$$

rectangular planform

$$AR = \frac{b^2}{S} = \frac{1600}{200} = 8$$

$$\Rightarrow c = 5 \text{ ft}$$

$$\mu(\alpha - \alpha_L) \sin \phi = A_1 \sin \phi (\mu + \sin \phi) + A_3 \sin 3\phi (3\mu + \sin \phi)$$

$$\phi = 45^\circ \quad \sin 45^\circ, A_1 \quad \sin 3\phi$$

$$0.19625 (0.06981) 0.70711 = 0.70711 (0.19625 + 0.70711) + A_3 (0.70711) (0.58875 + 0.70711)$$

$$0.00969 = 0.63878 A_1 + 0.91632 A_3 \quad (1)$$

$$\phi = 90^\circ$$

$$0.19625 (0.06981) = A_1 (1 + 0.19625) + A_3 (0.58875 - 1)$$

$$0.013700 = 1.19625 A_1 + 0.41125 A_3 \quad (2)$$

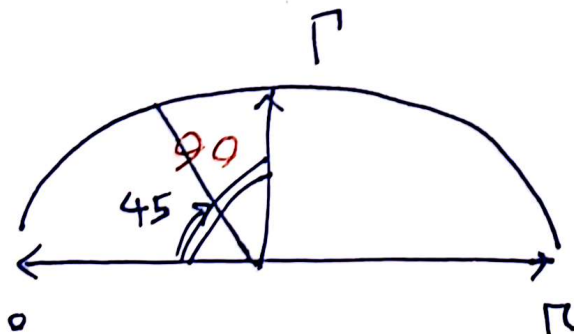
$$\textcircled{1} \text{ and } \textcircled{2} \Rightarrow A_1 = 0.010281$$

$$A_3 = 0.003408$$

$$C_L = RAR A_1 = 0.258$$

$$C_D = \frac{CL^2}{RAR} \left[1 + 3 \left(\frac{A_3}{A_1} \right)^2 \right] = 0.0035$$

"Always check"

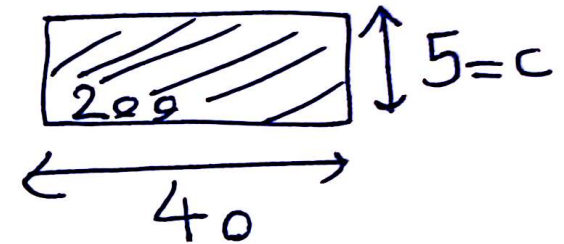


$$A_n > A_{n+1}$$

or

$$\frac{A_{n+1}}{A_n} < 1$$

Prandtl's Lifting Line Theory – Practice Problems



1. A wing has a planform area S of 200 ft^2 and a total span b of 40 feet. The airfoils are symmetric all along the span. The airfoil has a 2-D lift curve slope of 2π per radian. The wing has a rectangular planform, and thus has zero taper. The wing is untwisted.

- a. Compute the lift coefficient C_L and the drag coefficient C_{Di} at an angle of attack of 4 degrees.

Use two terms in the series expansion for circulation. $\int$

$$\Gamma = 2bV_\infty [A_1 \sin \phi + A_3 \sin 3\phi]$$

You may compare your results against the output of the Matlab code `prandtl.m` as we will explain in the class.

- b. Repeat the above calculation, now with just one term $\Gamma = 2bV_\infty A_1 \sin \phi$. Compare the lift drag coefficient C_L and C_D values to problem #2 above.
- c. Compare the results for drag coefficient from part (b) above with that for an elliptically loaded wing at this lift coefficient.

$$a_0 = 2\pi$$

$$2b = 45$$